A common action for a group and its automorphisms

Human Compression project

The acquired regular-polytope enumerator requires permutation groups $G \triangleleft A$ in the same action, with $|A : C_A(G)| = |\operatorname{Aut}(G)|$. A separately named permutation model of $\operatorname{Aut}(G)$ does not by itself supply this input. We use the following elementary construction to connect already identified models to the acquired enumerator.

Proposition 1. *Let H be a finite permutation group and $a = |\operatorname{Aut}(H)|$. Either of the following constructions supplies a faithful copy G of H and a conjugating group A inducing every automorphism of G.*

1. *Begin with $A = H$ in its given action. Adjoin permutations which normalize H. Accept this action when $|A : C_A(H)| = a$ and put $G = H$.*

2. *Let I be the complete set of involutions of H. Let G be the image of the conjugation action of H on I, and let A be the image of the action of $\operatorname{Aut}(H)$ on I. Accept these actions when both are faithful. Then $G \triangleleft A$ and $|A : C_A(G)| = a$.*

Proof. In the first construction, the supplied permutations normalize H, so the resulting group A does too. Conjugation gives a homomorphism $A \to \operatorname{Aut}(H)$ with kernel $C_A(H)$. Equality of the finite orders implies that its image is all of $\operatorname{Aut}(H)$.

For the second construction, automorphisms preserve element order, so I is invariant under both actions. Write $\iota : H \to \operatorname{Sym}(I)$ and $\alpha : \operatorname{Aut}(H) \to \operatorname{Sym}(I)$ for their homomorphisms. Their images are G and A. Faithfulness of ι identifies H with G; this is a checked condition, not an assumption based on the group name. For $h \in H$ and $\phi \in \operatorname{Aut}(H)$, the conjugation actions satisfy

$$\alpha(\phi)^{-1}\iota(h)\alpha(\phi) = \iota(\phi(h)),$$

with permutations acting on the right. Hence G is normal in A, and the induced automorphisms of G are precisely the transports of all automorphisms of H. If $\alpha(\phi)$ centralizes G, the displayed equality and injectivity of ι give $\phi(h) = h$ for every h. Thus $C_A(G)$ is trivial and $|A : C_A(G)| = a$. □

In the implementation, a proposed normalizing permutation is tested on every generator of H. Inclusion $H^c \leq H$ implies equality because conjugation preserves the finite order. Candidate overgroups are only a way of obtaining a smaller faithful action; their names and orders are never evidence for the automorphism condition. If none suffices, the implementation forms I from all involution conjugacy classes and checks its cardinality against their sizes. The orders of both action images certify faithfulness. It additionally checks normality and the centralizer quotient directly.

The generator images of H under ι retain the connection to the original named model. The acquired enumerator then applies unchanged to (G, A), and an isomorphism preserves generation, string relations, all subset intersections and equivalence under automorphisms. Its ordered classifications therefore transfer to H. Completeness still requires a complete returned enumeration; successful action setup alone is not a polytope census.

The use of involutions as a common set is borrowed from the already reviewed exceptional Coxeter computations in F006. The enumeration, intersection criterion, rank bound and exact GAP operations are acquired from the reviewed L071 application. This component provides the compatibility argument and explicit checked interface, not a new classification theorem or an open-conjecture solution.